

Opisanie funkcij szlzhivaniya

(Gitridnaya versiya: matematika + Python)

1. Ravnosnoe skolzhashee srednee (Boxcar)

Matematicheskaya formula

```
start(i) = max(0, i + 1 - m)
end(i)    = min(m + i, N)
W(i)      = end(i) - start(i)

y[i] = (1 / W(i)) * sum(x[j], j=start(i)..end(i)-1)
```

$m = M // 2$ — polovina okna. Na granicah okno obrezajetsya (clamping).

Realizacija — Python

```
def boxcar(win, data):
    half = win // 2
    n = len(data)
    avg = []
    for i in range(n):
        start = max(0, i + 1 - half)
        end = min(half + i, n)
        sum_array = sum(data[start:end])
        aver = sum_array / len(data[start:end])
        avg.append(aver)
    return avg
```

Svojstva

- Vesa: ravnye $1/W(i)$
- ACH: $H(f) = \sin(\pi f M) / (M \cdot \sin(\pi f))$
- Slozhnost': $O(N \cdot \text{win})$

2. Gaussov filtr (Gaussian)

Matematicheskaya formula

```

σ = M / 6
w[k] = exp( -(k - m)2 / (2·σ2) ), k = 0..M-1
w_norm[k] = w[k] / cyμμα(w)

y[i] = cyμμα( w_norm[k] · x_padded[i+k], k=0..M-1 )

```

Signal dopolnjaetsja simmetricheskim otrazheniem (mirror reflection) na m elementov s kazhdoj storony.

Realizacija — Python

```

def gaussian(data, win):
    half = win // 2
    sigma = win / 6.0
    n = len(data)
    k = [math.exp(-0.5*(i - half)**2 / sigma**2)
          for i in range(win)]
    k = [x/sum(k) for x in k]
    p = ([data[i] for i in range(half-1,-1,-1)]
          + data
          + [data[i] for i in range(n-1,n-half-1,-1)])
    return [sum(k[j]*p[i+j] for j in range(win))
            for i in range(n)]

```

Svojstva

- Vesa: ulyvajut po Gussu, net artefaktov Gibbsa
- Simmetrichnost' yadra: nulevaja faza
- $\sigma = M/6$: ~99.7% energii vnutri okna
- Slozhnost': $O(N \cdot \text{win})$

3. Progressivnoe srednee (Progressive)

Matematicheskaja formula

```
y[i] = (1/(i+1)) · cyμμα( x[k], k=0..i )
```

Rekurrentno:

```

S[i] = S[i-1] + x[i]
y[i] = S[i] / (i + 1)

```

Realizacija — Python

```

def progressive(data):
    result = []
    total = 0.0
    for i, value in enumerate(data):
        total += value
        result.append(total / (i + 1))
    return result

```

Svojstva

- Razmer okna: i+1 (rastot ot 1 do N)
- Dispersija: $O(1/i)$ — minimum na poslednem otsc'jote

- Slozhnost': $O(N)$

4. Gibridnoe szlzhivanie (Hybrid)

Matematicheskaja formula

```

y[i] =
    y_boxcar[i],                esli 0 <= i < M
    (1-λ) · y_boxcar[i] + λ · y_prog[i],    esli M <= i < 2M
    y_prog[i],                  esli i >= 2M

λ(i) = (i - M) / M,  λ  [0, 1)

```

Realizacija — Python

```

def hybrid(data, win):
    n = len(data)
    box = boxcar(win, data)
    prog = progressive(data)

    out = []
    for i in range(n):
        if i < win:
            out.append(box[i])
        elif i < 2 * win:
            z = (i - win) / win
            k = (1.0 - z) * box[i] + z * prog[i]
            out.append(k)
        else:
            out.append(prog[i])
    return out

```

Zony

Zona	Diapazon	Formula	Povedenie
I	$0 \leq i < M$	$y = y_{\text{boxcar}}$	Stabil'noe usrednenie
II	$M \leq i < 2M$	$(1-\lambda) \cdot \text{box} + \lambda \cdot \text{prog}$	Linejnyj crossfade
III	$i \geq 2M$	$y = y_{\text{prog}}$	Minimal'naja dispersija

Svojstva

- Nepreryvnost': garantirovana na vseh granicah
- Perehodnaja zona: shirina M otc'jotov
- Szlzhivanie: monotonno usilivaetsja ot I k III